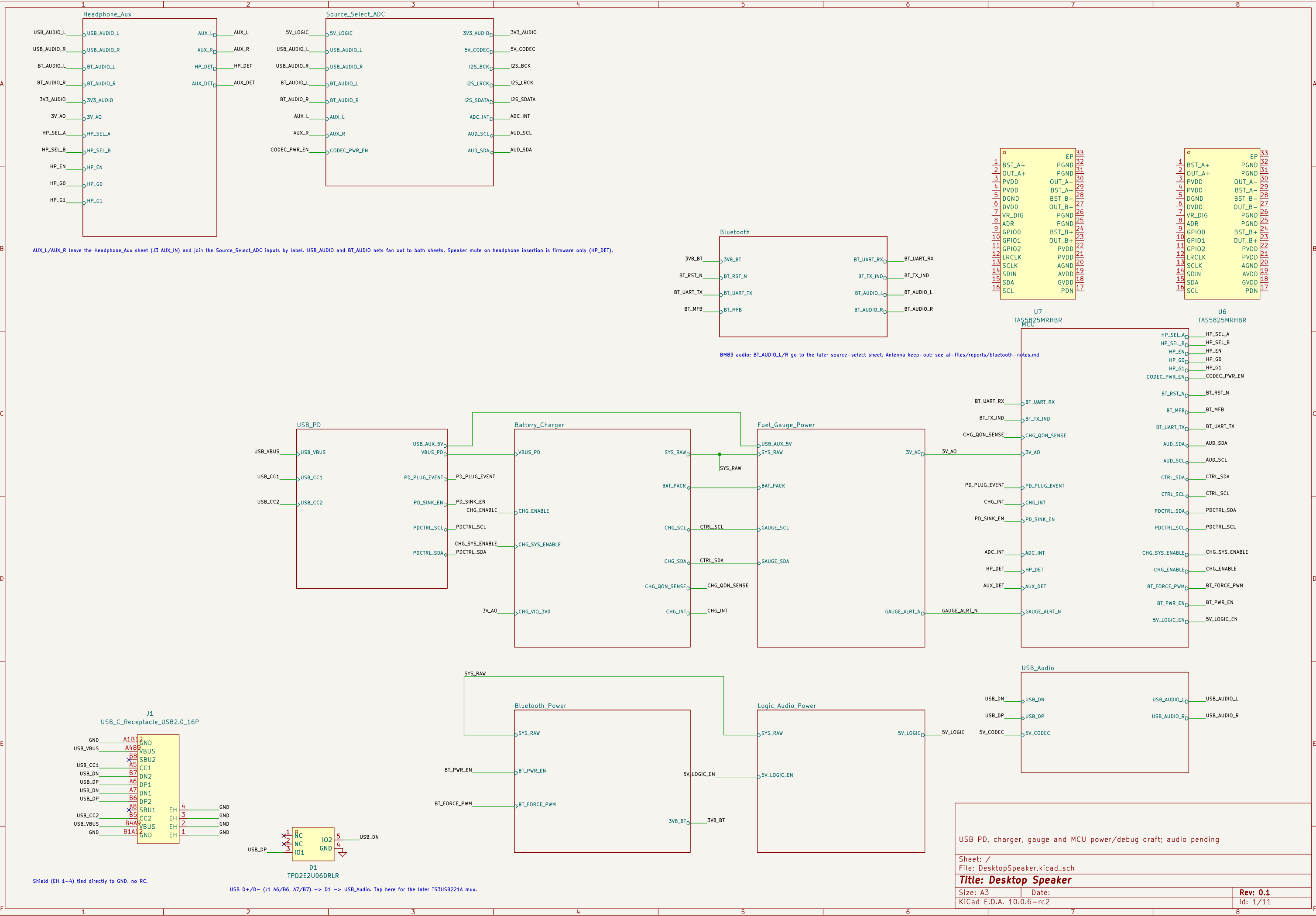
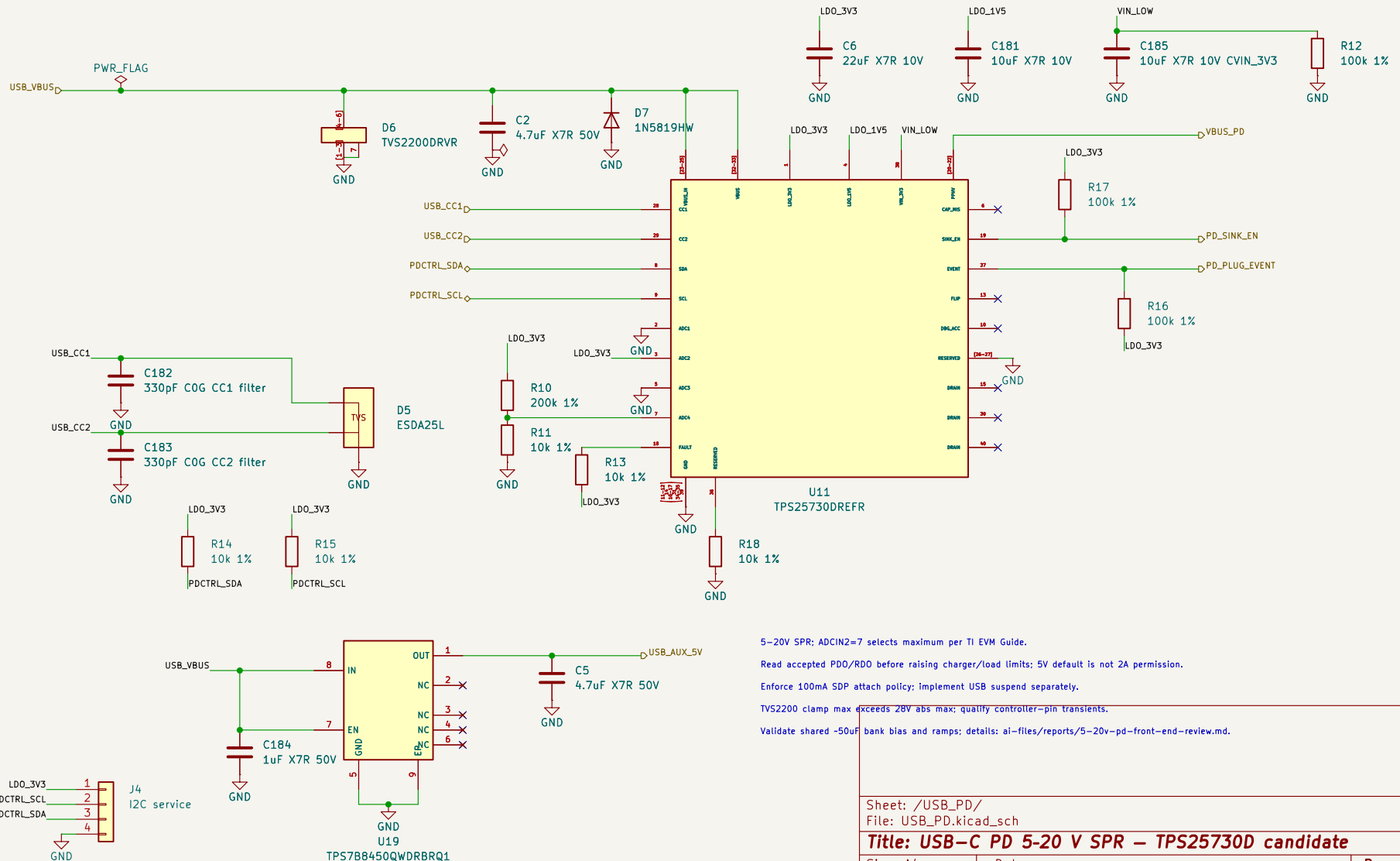
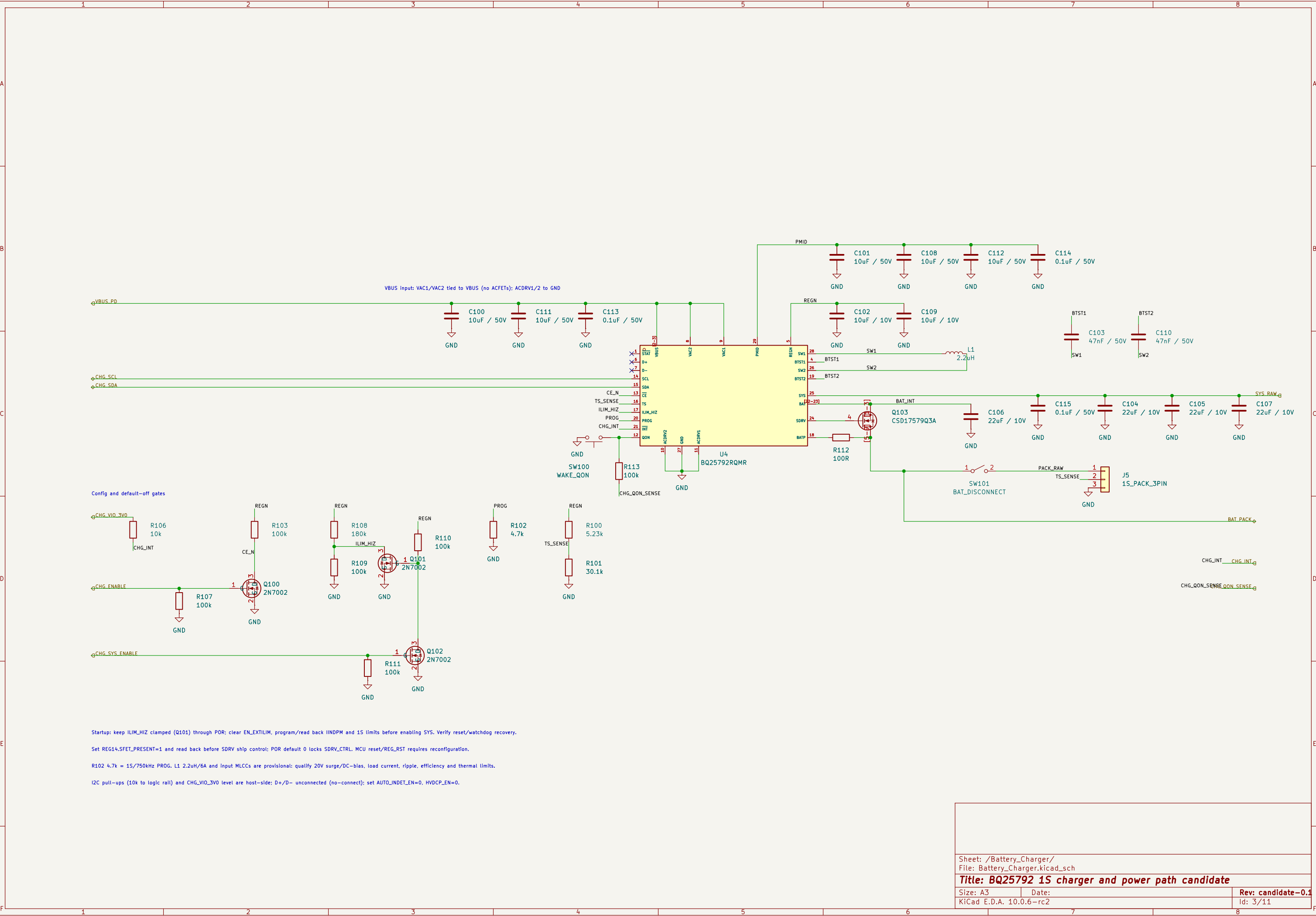
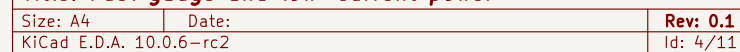
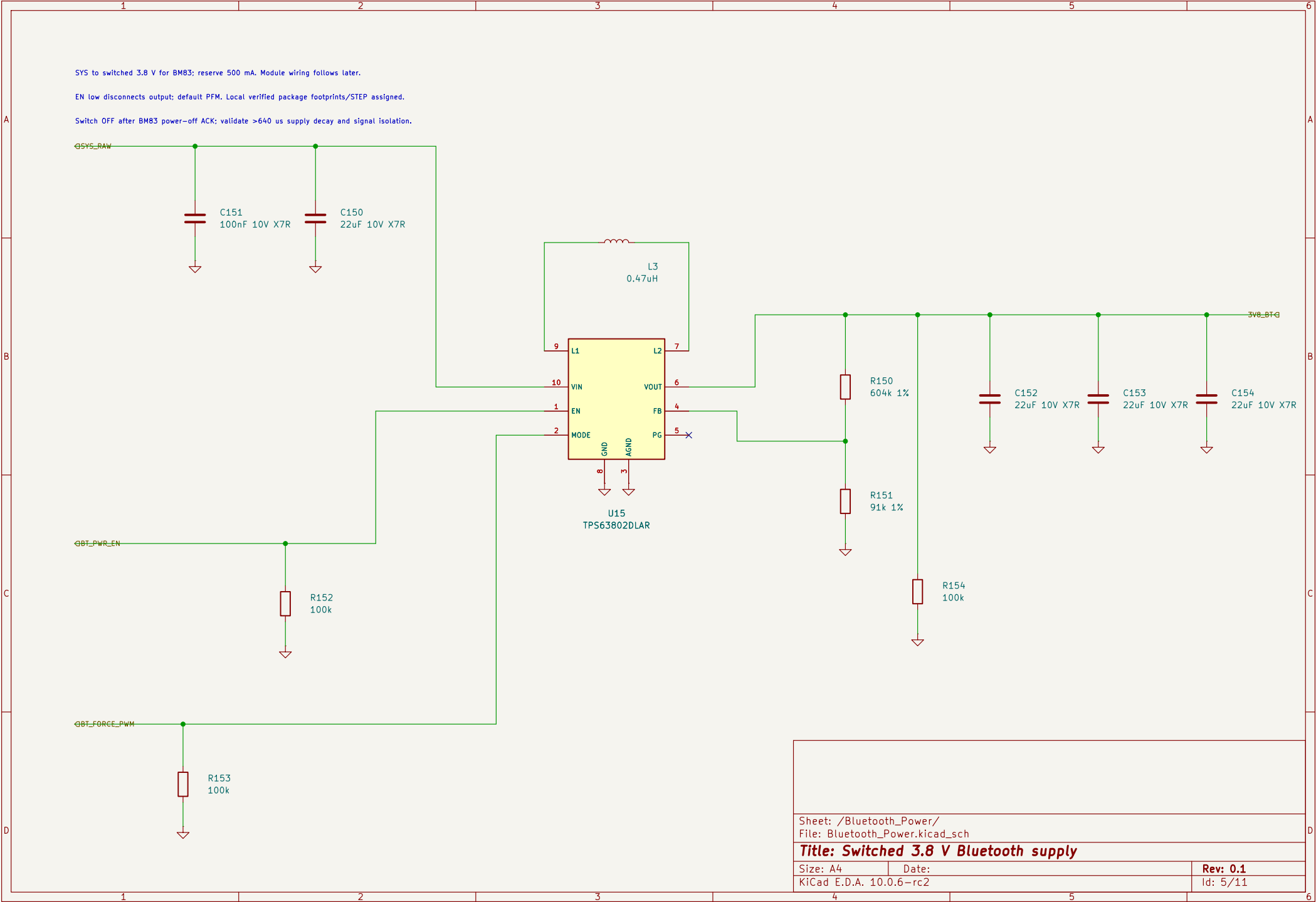








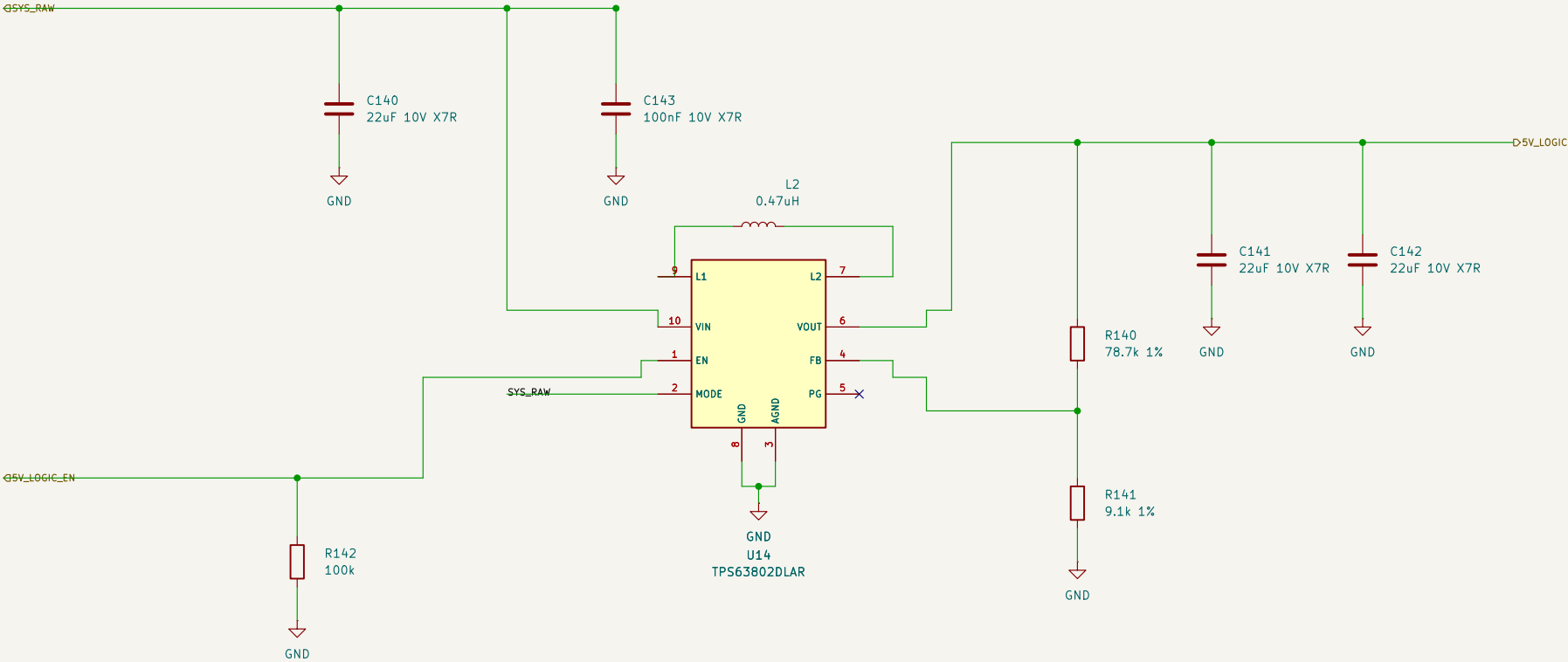




Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

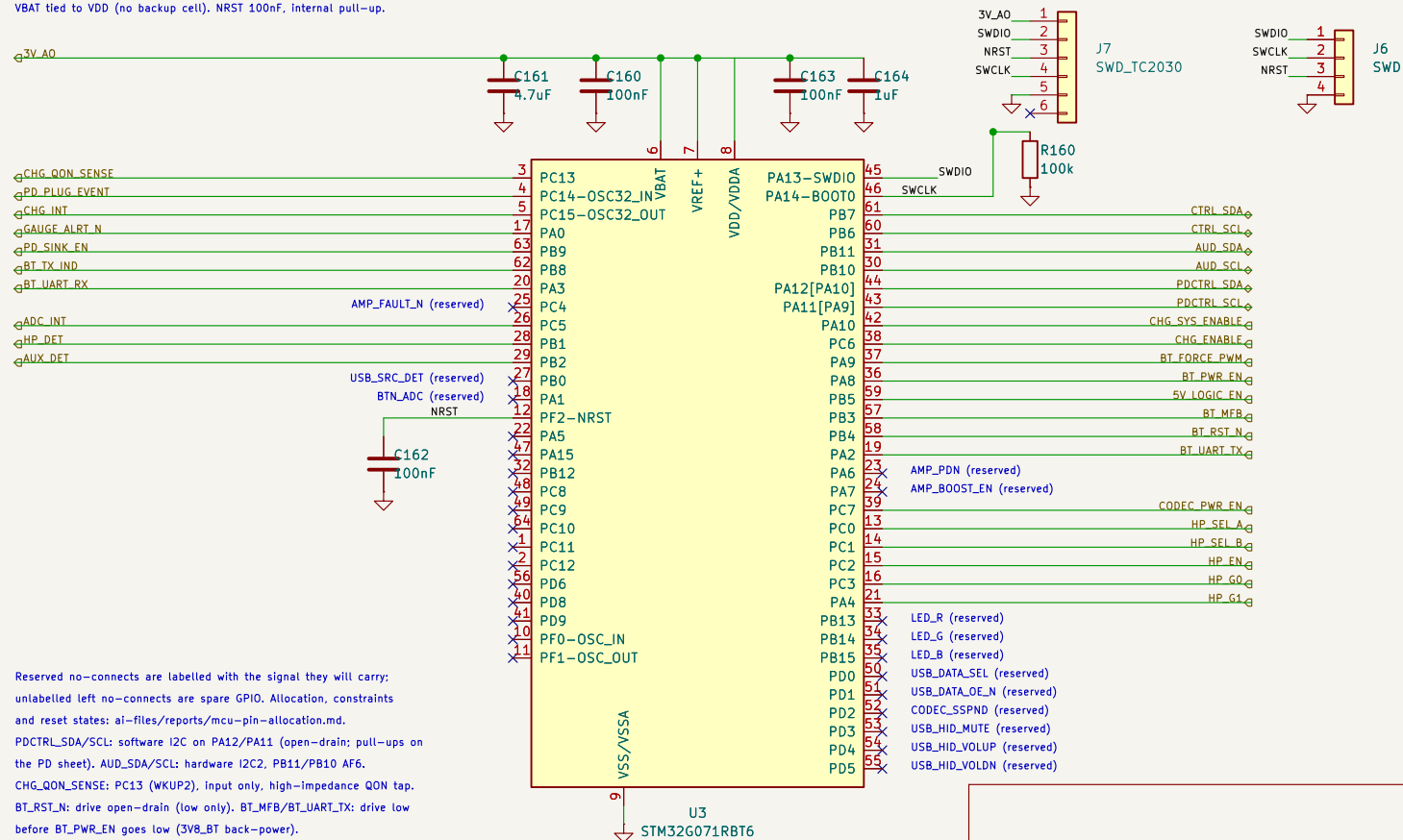
EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_AO supply: VDD/VDDA 100nF + 4.7uF (DS12232 fig. 13). VREF+ 100nF + 1uF (VREFBUF off, VREF+ = VDDA).
VBAT tied to VDD (no backup cell), NRST 100nF, internal pull-up.

SWD: J7 Tag-Connect TC2030-IDC (with legs: 6 SWO unused) and J6 4-pin 2.54 mm header share SWDIO, SWCLK, NRST
R160 holds PA14/BOOT0 low (nBOOT_SEL=1 ignores the pin)



STM32G071RBT6 LQFP64: supply, reset, SWD, power-block links

Sheet: /MCU/
File: MCU.kicad_sch

Title: MCU and programming

Size: A4

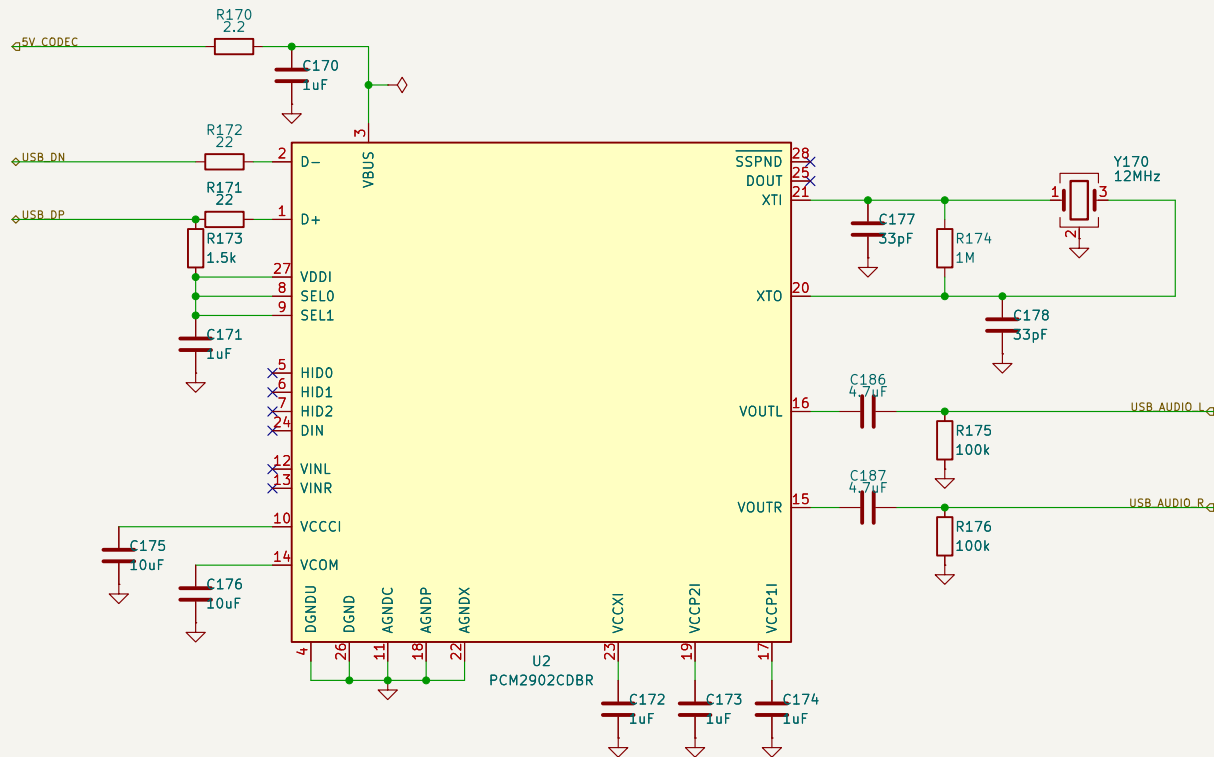
Date:

Rev: 0.1

KiCad E.D.A. 10.0.6-rc2

Id: 7/11

Supply: 5V_CODEC (U23 load switch from 5V_LOGIC, 4.60–5.06 V worst case) feeds VBUS through 2.2R/1uF. DS range 4.35–5.25 V, 56 mA typ / 67 mA max, 250 uA suspend.
 Codec is powered only when CODEC_PWR_EN (U23) and 5V_LOGIC are on: D+ pull-up (R173 to VDDI) = USB attach. Do not enable CODEC_PWR_EN on DCP/unknown sources.
 SEL0/SEL1 = high (VDDI). HID0–2, DIN, VINL/R unused. SSPND (suspend flag, active low) reserved for a later MCU input.
 USB_DP/USB_DN come from J1 via D1 (root). Later source-detect mux goes between those root nets and these ports.



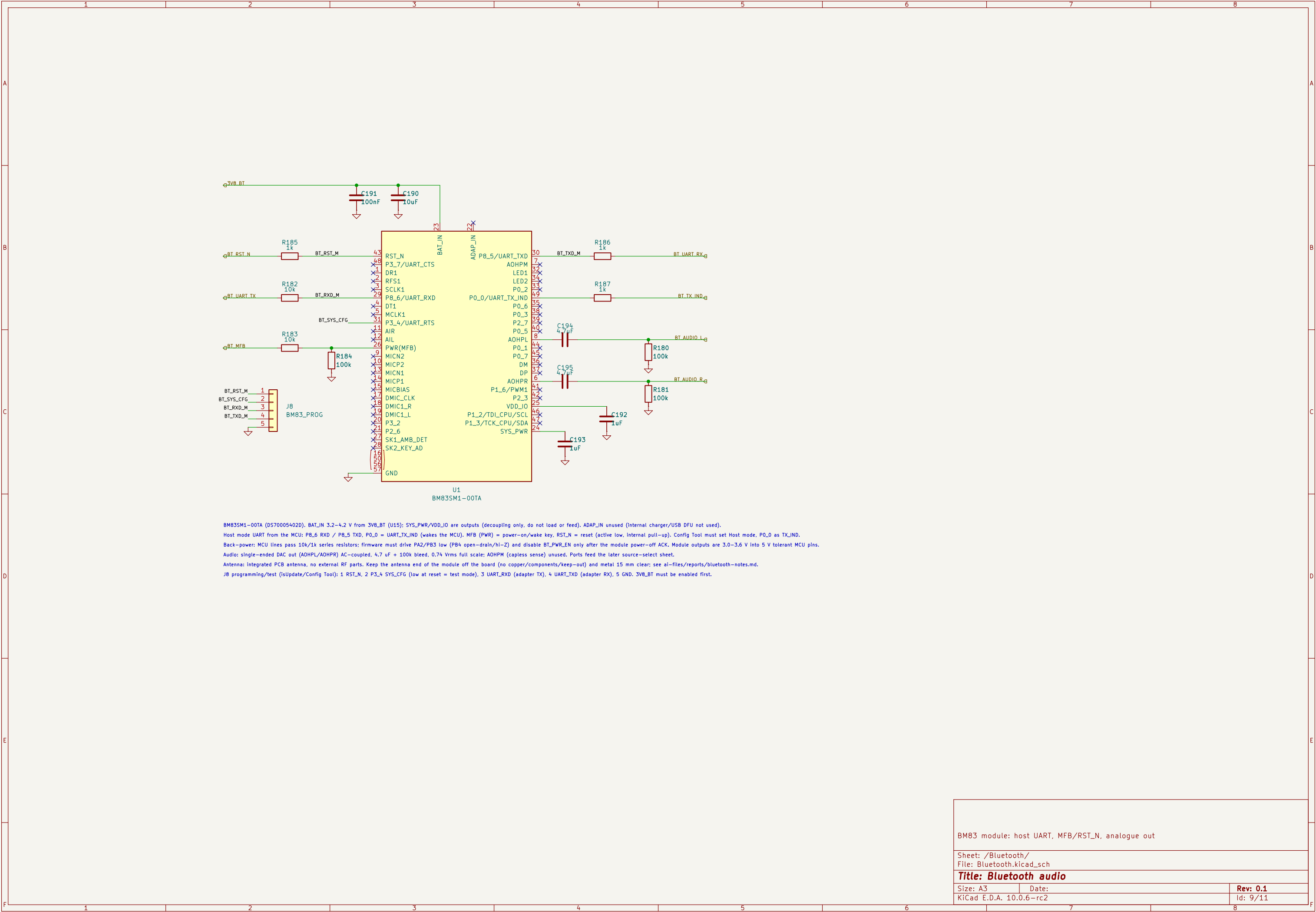
Analogue out: 0.6 VCCCI Vpp (about 2.0 Vpp, 0.7 Vrms) centred on 1.65 V; AC-coupled (HPF -0.4 Hz at 100k), >=10k load.

PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D- path; analogue out to USB_AUDIO_L/R

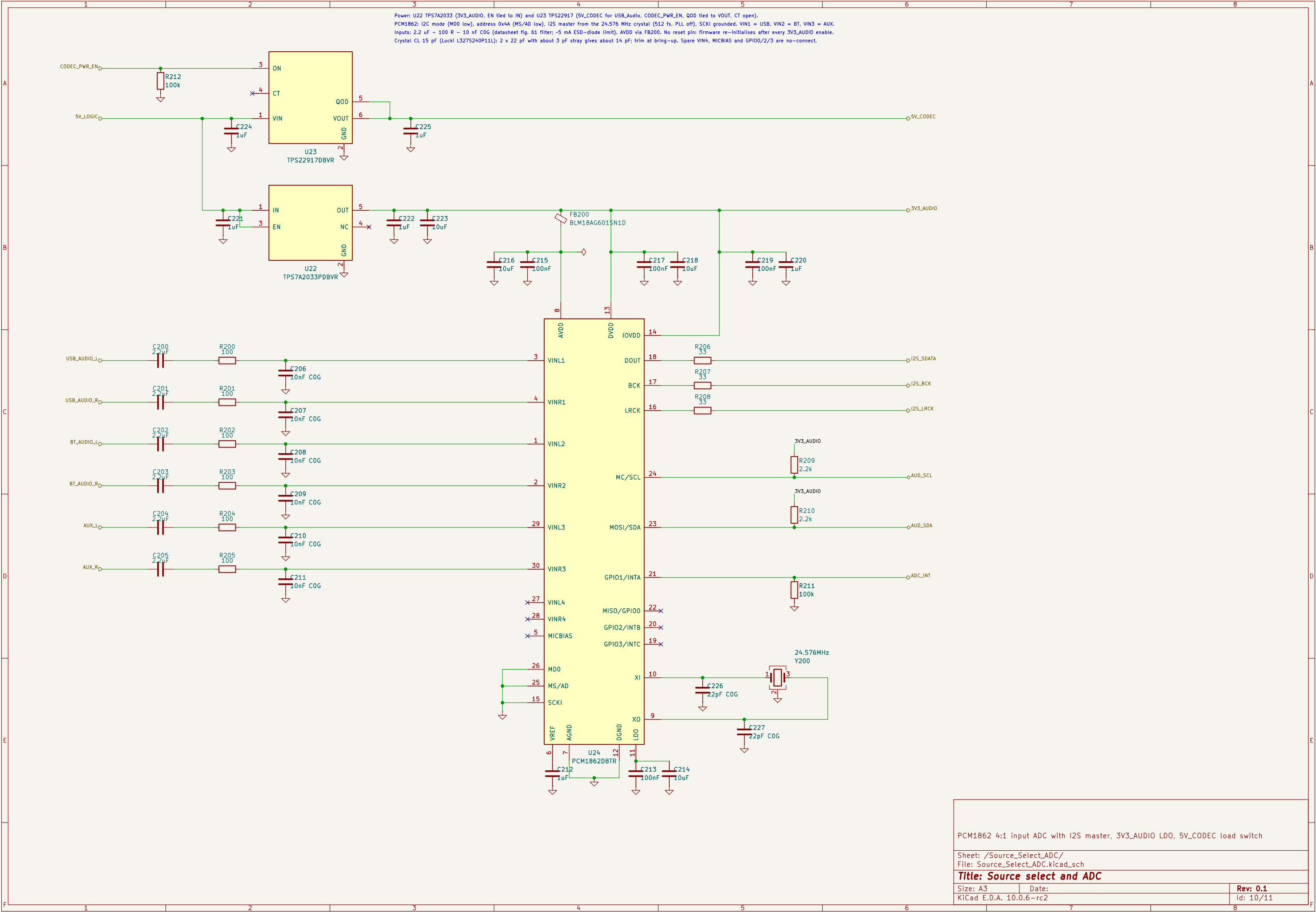
Sheet: /USB_Audio/
 File: USB_Audio.kicad_sch

Title: USB audio

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 8/11



BM83 module: host UART, MFB/RST_N, analogue out		
Sheet: /Bluetooth/ File: Bluetooth.kicad_sch		
Title: Bluetooth audio		
Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 9/11



PCM1862 4:1 input ADC with I2S master, 3V3_AUDIO LDO, 5V_CODEC load switch

Sheet: /Source_Select_ADC/
File: Source_Select_ADC.kicad_sch

Title: Source select and ADC

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 10/11

